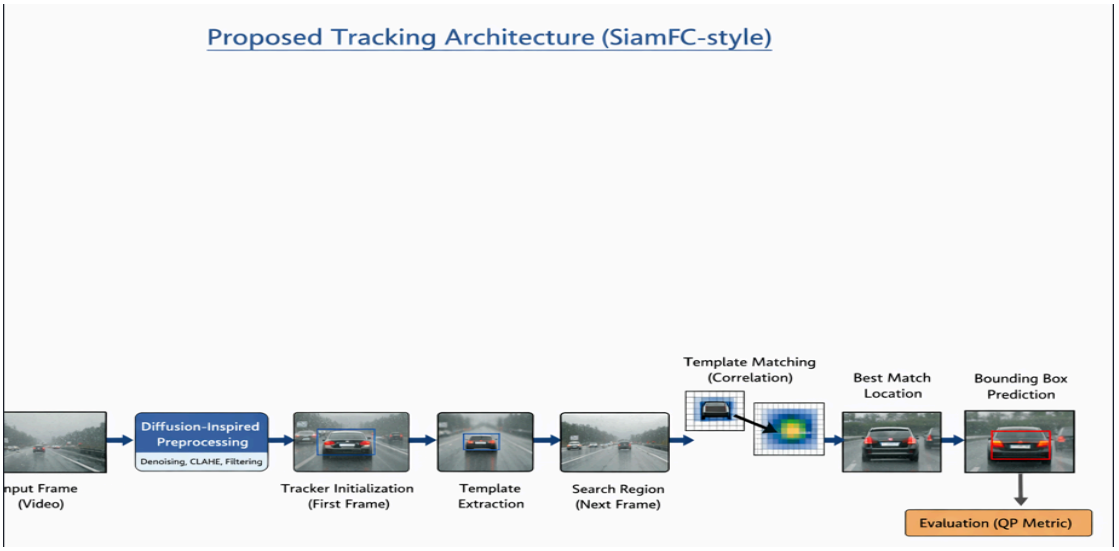
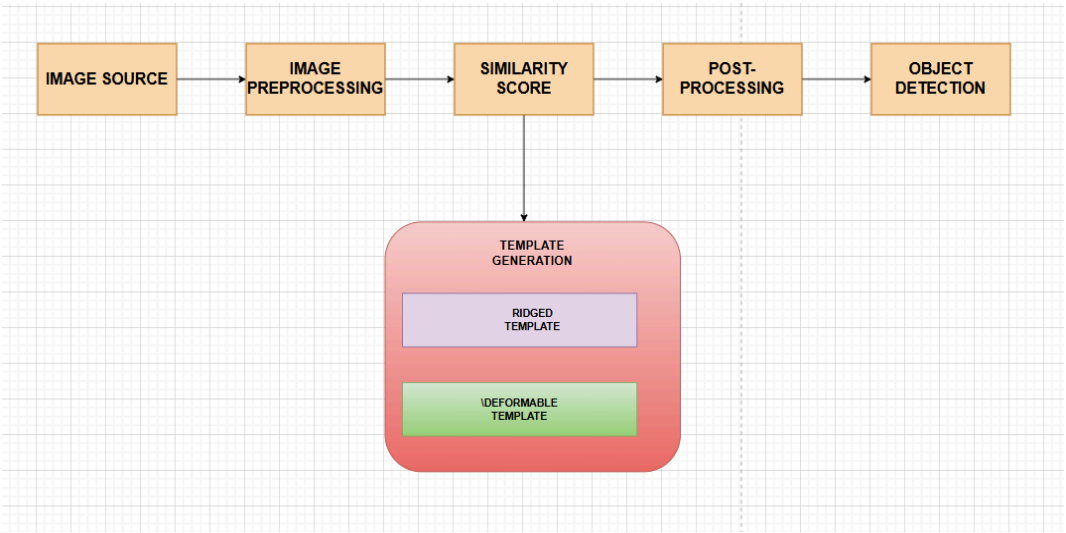
MODEL ARCHITECTURE

“We designed a SiamFC-style tracking pipeline where similarity between template and search region is used for tracking. Instead of a deep CNN, we used a lightweight template matching approach.”





Proposed SiamFC-style Tracking Pipeline

Step 1: Input Video Frame

The system takes video frames as input from the dataset (rainy or hazy conditions).

Step 2: Diffusion-Inspired Preprocessing

Each frame is enhanced using preprocessing techniques such as:

- Noise reduction
- Contrast enhancement (CLAHE)
- Filtering
  This step improves visibility under rain and haze conditions.

Step 3: Tracker Initialization (First Frame)

In the first frame:

- Ground truth bounding box is used
- Tracker is initialized with object location

Step 4: Template Extraction

The object region from the first frame is extracted as a template.
This template represents the target to be tracked.

Step 5: Search Region Selection (Next Frames)

For each new frame:

- A local search region is selected around the previous object position
- This reduces computation and improves speed

Step 6: Template Matching (Correlation)

The template is matched with the search region using:

- Correlation
- Similarity is computed to find best match

Step 7: Best Match Location

The location with the highest similarity score is selected as:

- The new object position

Step 8: Bounding Box Prediction

A bounding box is generated at the detected location:

- Same size as template (or adjusted)
- Represents predicted object position

Step 9: Repeat for All Frames

Steps 5–8 are repeated for all frames in the sequence to continuously track the object.

Step 10: Evaluation (QP Metric)

Tracking performance is evaluated using:

- QP (Quality Precision) metric
- Measures distance between predicted and ground truth centers